

CHEM 101 Topic 7: Chemical Reactions — Answer Key

Section 1: Writing and Balancing Equations

Problem 1: Reactants: Methane (CH₄), Oxygen (O₂); Products: Carbon dioxide (CO₂), Water (H₂O).

Problem 2: $2\text{Al} + 3\text{CuCl}_2 \rightarrow 2\text{AlCl}_3 + 3\text{Cu}$

Problem 3: $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$

Section 2: Types of Chemical Reactions

Problem 4: Combination (or Synthesis) reaction.

Problem 5: $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Problem 6: (a) Single Replacement; (b) Double Replacement.

Section 3: Solubility Rules & Precipitation

Problem 7: No precipitate; both NaCl and KNO₃ are soluble.

Problem 8: PbCl₂(s).

Problem 9: $3\text{AgNO}_3(\text{aq}) + \text{K}_3\text{PO}_4(\text{aq}) \rightarrow \text{Ag}_3\text{PO}_4(\text{s}) + 3\text{KNO}_3(\text{aq})$

Section 4: Clinical Applications & Synthesis

Problem 10: Double-replacement reaction.

Problem 11: Ca₃(PO₄)₂.

Problem 12: $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$